SOC-5811 Week 15: What have we learned?

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Review: Big Picture

What is the point of building a model?

- **Prediction**

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- **Prediction**
- **Comparison**

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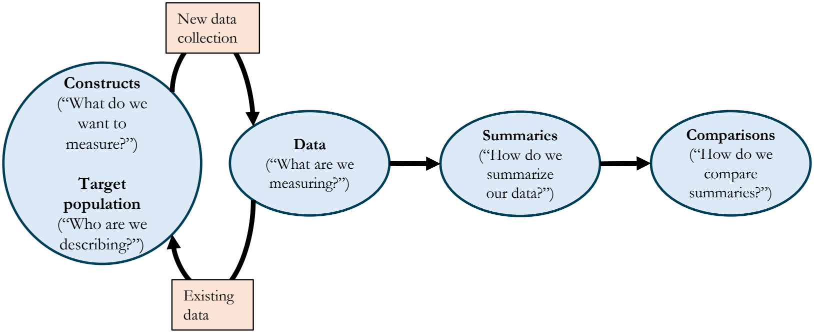
- Regression coefficients can always be interpreted as average comparisons of predictions.
- Regression coefficients can sometimes be interpreted as effects.

Review: Big Picture



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  - **Constructs:** Do our measures generalize to concepts?
  - **Target population:** Does our sample generalize to the population?
  - **Forecasting:** Does our model of the present generalize to the future?
  - **Causal inference:** Does our comparison generalize to a causal effect?

Review: Big Picture

What do we need in order to interpret our comparison as a causal effect?

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- **Internal validity** (e.g., exchangeability)
- **External validity** (e.g., transportability)

Review: Models

What does it mean to "build" a model?

Review: Models

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- **Estimand:** What is my goal? What is my target comparison I want to infer something about? Example: I want to know if income varies by education, conditional on age.
- **Estimator:** How do I relate my population estimand to real sample data in a way that let's me identify its *most likely* value given that sample data?
- **Estimation:** How do I optimize my estimator (i.e., calculate the most likely value of my estimand)?

Review: Models

1. **Estimand:** Choose a functional form for the model and highlight the target parameter ("estimand").

$$y_i = \beta_0 + \boxed{\beta_1} x_i + \epsilon_i$$

2. **Estimator:** Choose an estimator that relates the model to real observed data (e.g., OLS) and is unbiased, consistent, and efficient.

$$\text{Sum of squared residuals} = \sum_{i=1}^{n} (y_i - \widehat{\beta_0} - \widehat{\beta_1} x_i)^2$$

3. **Estimation:** Estimate parameters by optimizing the estimator; sometimes this can be done analytically, sometimes it requires brute computational force.

Review: Statistical Inference

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- We often want to do **statistical inference** on these parameters because they represent comparisons that map to our research questions.
- This involves **uncertainty**.

Review: Statistical Inference

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- How do we estimate the standard error?
  - Analytic approach: formulas.
  - Simulation approach: bootstrapping.

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- For inference, we use the **p-value:** the probability that the null model would produce a parameter estimate as extreme as your observed estimate if the null model is correct.
- Our convention is that a p-value of less than 0.05 represents "statistical significance."

Review: Causal Inference

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  - We want to estimate a causal effect, so we control for confounders.

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  - We can only ever see one outcome at one time for any given unit, even though they have multiple *potential outcomes*.
- We can attempt to estimate **average treatment effects** by predicting potential outcomes.

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- The regression model is one way of generating the predictions needed to estimate the ATE.

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- **Mediator:** any variable on the causal pathway between treatment and outcome.
- **Moderator:** any variable that affects the relationship between treatment and outcome.

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- **Panel data:** A data set where we observe the same units (e.g., individuals, firms, countries, schools) over more than one time period.
- New estimator: fixed effects.
- Fixed effects for the unit of analysis removes time-invariant confounding.
- Do I care about within- or between-unit variation for my research question?

Review: Assumptions

Key assumptions:

- Validity
- Representativeness
- **Additivity and linearity**
- Independence of errors
- Equal variance of errors
- Normality of errors

Review: Assumptions

- Linear transformations

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  - Shifting (i.e., add/subtract a constant)
  - Rescaling (i.e., multiply/divide by a constant)

Review: Assumptions

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- **Causal model**
  - Examining heterogeneous treatment effects (transform/interact the treatment variable).
  - Relaxing assumptions about confounding pathways (transform/interact the confounder variables).

- Steps for an applied quantitative research project

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Applied Quantitative Research

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APPLIED QUANTITATIVE RESEARCH

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- **(2)** Familiarize yourself with the data. Can we use existing data? Do we need new data?
- **(3)** Refine conceptual model into a DAG (this may include drawing the "causal field") that is specific about measurement of *constructs* and definition of *target population*.
- **(4)** Specify the *estimand* that answers your research question. In other words, choose a functional form for the model and highlight your target parameter (e.g., β_1).

- **(5)** Load data into R. Make descriptive summary statistics and plots. Revisit Steps 1-4 as needed.

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- **(6)** Choose estimator (e.g., OLS) and fit model.
- **(7)** Test for non-linearity and non-additivity in the relationship between treatment and outcome.

Review

- Wednesday (12/10): What have we not learned?

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- Please complete the Student Rating of Teaching on the menu on the left of Canvas (due 12/10).

Pick up here Wednesday.

Why should we understand quantitative methods (and coding)?

- **There is some inherent value.** Quantitative methods can provide empirical, falsifiable facts about the social world (but not what those facts mean!); for example, how did different groups vote in 2024, and was this different from how those groups voted in 2020?

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- **Non-academic careers.**
- **Organizing/advocacy.**

Quantitative methods

- **Empiricism:** social theories imply empirical claims that are falsifiable.

QUANTITATIVE METHODS

- **Empiricism:** social theories imply empirical claims that are falsifiable.
- Data doesn't speak for itself; it must be carefully interpreted, summarize, and analyzed.

Review

Spirling & Stewart (2025): Inference to best explanation (IBE).

"Research contributions therefore fall into three overlapping types: the generation of explanations, the production of facts, and the updating of explanations in light of those facts."

Review

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- Second, explanation is obviously vital to the practice of IBE, and should be taken as seriously as inference is.

- Third, that exploratory analysis, because it encourages the development of explanations, is similarly crucial and currently undervalued."

TheUpshot

The Americans Most Threatened by Eviction: Young Children

About a quarter of Black babies and toddlers in rental households face the threat of eviction in a typical year, a new study says, and all children are disproportionately at risk.

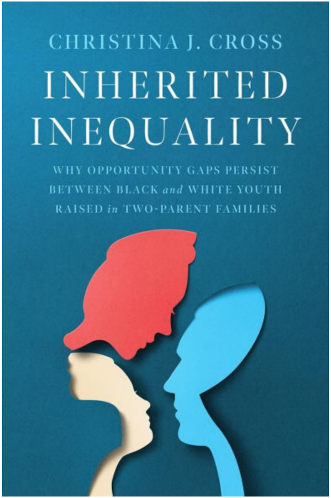
Share full article 141

By Emily Badger, Claire Cain Miller and Alicia Parlapiano

Oct. 2, 2023

The Americans most at risk of eviction are babies and toddlers, according to new data that provides the fullest demographic picture yet of who lives in rental households facing eviction nationwide.

Review


CHRISTINA J. CROSS
INHERITED INEQUALITY
WHY OPPORTUNITY GAPS PERSIST BETWEEN BLACK and WHITE YOUTH RAISED in TWO-PARENT FAMILIES

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WHAT HAVE WE NOT LEARNED?

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    - Advanced **confounder control** methods (e.g., matching, weighting, g-methods).

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- Non-linear regression models (i.e., generalized linear models, or GLMs) (**SOC 8811**)
- Causal inference (**SOC 8890**)
  - Advanced **confounder control** methods (e.g., matching, weighting, g-methods).
  - Advanced **instrument-based** methods (e.g., IVs, diff-in-diff).

What have we not learned?

- Statistical inference

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- Survey design

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- Statistical inference
- Survey design
- Advanced predictive models (i.e., "machine learning")
- Free online resources outside of classes

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Introduction to the course

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INTRODUCTION TO THE COURSE

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- Norms, computation standards, etc.
- Generating knowledge is hard no matter what methods we use to do it.